RIGIDITY: THE RELATIVE DUALIZING COMPLEX INVERTED ALONG THE DIAGONAL GIVES BACK THE STRUCTURE SHEAF AND THE COMPLEX

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E2P (sources/stacks-duality/pinned-original-source.tex line 3159).

Allowed standard prerequisites (editorial).

- The right adjoints of pushforward and their composition along the diagonal. Same chapter, Lemma labelled lemma-twisted-inverse-image, cited at proof lines 3225-3235. Define a, b, c and the identity id = c o b.
- The comparison with pullback for a flat proper morphism. Same chapter, Lemma labelled lemma-compare-with-pullback-flat-proper, cited at proof lines 3237-3240. Gives the second term of the chain.
- Base change of the adjoint along the projection. Same chapter, Lemma labelled lemma-proper-flat-base-change, cited at proof lines 3240-3244. Gives the third term.
- The relative dualizing complex as the adjoint applied to the structure sheaf. Same chapter, Remark labelled remark-relative-dualizing-complex, invoked in the statement, used at proof lines 3244-3248. Identifies a(O_Y) with omega.

Source reference. Original reference 0B6S: sources/stacks-duality/pinned-original-source.tex, source lines 2945–2945. Same-chapter pointer into the author environment remark at source line 2945; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 3186 of the counted proof. It is

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Source reference. Original lemma 0E4K: sources/stacks-duality/pinned-original-source.tex, source lines 2892–2892. Same-chapter pointer into the author environment lemma at source line 2892; invoked at proof line 3217 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AAB: sources/stacks-duality/pinned-original-source.tex, source lines 2929–2929. Same-chapter pointer into the author environment lemma at source line 2929; invoked at proof line 3218 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1 (Rigidity)**.** *Let Y be a quasi-compact and quasi-separated scheme. Let $f : X \to Y$ be a proper, flat morphism of finite presentation with relative dualizing complex $\omega_{X/Y}^\bullet$ (Remark 0B6S). There is a canonical isomorphism*

$$\mathcal{O}_X = c(Lpr_1^*\omega_{X/Y}^\bullet) = c(Lpr_2^*\omega_{X/Y}^\bullet) \tag{0.1.1}$$

and a canonical isomorphism

$$\omega_{X/Y}^\bullet = c\left(Lpr_1^*\omega_{X/Y}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_Y X}} Lpr_2^*\omega_{X/Y}^\bullet\right) \tag{0.1.2}$$

where c is the right adjoint of Lemma 0A9E for the diagonal $\Delta : X \to X \times_Y X$.

Proof. Let a be the right adjoint to Rf_* as in Lemma 0A9E. Consider the cartesian square

$$\begin{array}{ccc} X \times_Y X & \xrightarrow{q} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle f} \\ X & \xrightarrow{f} & Y \end{array}$$

Let b be the right adjoint for p as in Lemma 0A9E. Then

$$\begin{aligned} \omega_{X/Y}^\bullet &= c(b(\omega_{X/Y}^\bullet)) \\ &= c(Lp^*\omega_{X/Y}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_Y X}} b(\mathcal{O}_X)) \\ &= c(Lp^*\omega_{X/Y}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_Y X}} Lq^*a(\mathcal{O}_Y)) \\ &= c(Lp^*\omega_{X/Y}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_Y X}} Lq^*\omega_{X/Y}^\bullet) \end{aligned}$$

as in (0.1.2). Explanation as follows:

(1) The first equality holds as $\mathrm{id} = c \circ b$ because $\mathrm{id}_X = p \circ \Delta$.
(2) The second equality holds by Lemma 0E4K.
(3) The third holds by Lemma 0AAB and the fact that $\mathcal{O}_X = Lf^*\mathcal{O}_Y$.
(4) The fourth holds because $\omega_{X/Y}^\bullet = a(\mathcal{O}_Y)$.

Equation (0.1.1) is proved in exactly the same way. □